Day 13 — Genome Annotation

BIOL 4810

2026-03-08

Day 13 — Genome Annotation + Species Identification

Goal: Use **Prokka** (Galaxy) to annotate a bacterial genome assembly, interpret the output files, extract the **16S rRNA** for a quick BLAST-based ID, and then use **TYGS** + **iTOL** to place your genome on a phylogenomic tree.
Submission: Fill in the answers + upload required screenshots/files below → click **Prepare submission preview** → click **Download as PDF (Print)**.

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0) Files you need today

You should already have (from Day 12): - SPAdes assembly contigs file (FASTA)

0A. Assembly file name (as it appears in Galaxy History):

Sample1_spadescontigs.fasta

Local folder (for Git later)

Create a folder inside your BIOL 4810 course folder, for example: - `module2_wgs/day13_prokka_tygs/`

0B. Local folder path/description:

Bioinformatics/biol4810/module02/annotation/

0C) Filter contigs for Prokka (≥500 bp)

Some downstream tools struggle with very short contigs or very large numbers of contigs.

In Galaxy: 1. Search for the tool **Filter FASTA** (it may appear as *Filter sequences by length*). 2. Input: your original SPAdes **contigs FASTA**. 3. Set **minimum sequence length = 500 bp**. 4. Run the tool to create a **filtered contigs** dataset.

Use the filtered contigs for Prokka today.

Keep the original (unfiltered) contigs for TYGS. TYGS can take a long time and you already submitted the full assembly there.

0C. Filtered contigs dataset name (Galaxy History):

Galaxy54-[Filter FASTA on dataset 52_ FASTA sequences]

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1) Genome QC (CheckM2 in Galaxy)

Today you will start by evaluating your assembly quality in **Galaxy** using **CheckM2**, which estimates **completeness** and **contamination**.

1A — CheckM2

In Galaxy: 1. Run **CheckM2** on the **filtered assembly FASTA** you will use for Prokka. 2. Open the main CheckM2 results table/report.

CheckM2 tool version (from Job Info):

(Galaxy Version 1.1.0+galaxy0)

Completeness (%):

98.96

Contamination (%):

0.1

Interpretation (2–3 sentences): Would you trust this assembly for annotation/species ID? Why?

I would trust this because of the high percentage of completeness and low percentage of contamination. Low contamination means that their sequence is mostly comprise from a single organism, and the high completeness indicates that the sequence contains most of the expected genes for the organism's genome.

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2) Run Prokka on Galaxy

In Galaxy: 1. Search tools for **Prokka** 2. Input: your SPAdes **contigs FASTA** 3. Use defaults unless instructed otherwise 4. Run Prokka

Record your Prokka settings

2A. Prokka version (from Job Info):

Version 1.14.6+galaxy1

2B. Any settings changed from defaults? If yes, what?

Force Genbank Compliance - YES

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3) Prokka outputs — what is each file for?

From Prokka outputs in Galaxy, you will use these most today:

- **.txt** summary statistics
- **.gff** master annotation (features + coordinates)
- **.ffn** nucleotide sequences of gene features (includes rRNA genes)
- **.faa** amino acid sequences of predicted proteins

3A. In your own words, what is the difference between `.gff`, `.ffn`, and `.faa`? (2–4 sentences)

.gff shows where genes and features are located on a genome by annotating where they are located (coordinates. .ffn contain the DNA sequences of genes. Finally, .faa shows the protein sequence that is predicted through the amino acid sequence.

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4) Interpret the Prokka summary (.txt)

Open the Prokka **.txt** summary.

4A. Organism line (paste exactly as reported — it may be "Unknown"):

Genus species strain

4B. contigs:

79

4C. bases:

4797060

4D. CDS:

4507

3E. rRNA (total rRNA features):

3

3F. tRNA:

62

3G. tmRNA:

1

3H. Based on genome size and CDS count, does this look like a typical bacterial genome? Why/why not? (2–3 sentences)

The CDS count falls within the range of a typical bacterial genome (1000-6000), and the genome size falls within the respective range for base pairs (1-6 million). With both of these taken into account, it appears that this looks like a typical bacterial genome.

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5) rRNA features (16S/23S/5S) — paste evidence from Prokka

The Prokka `.txt` summary often reports only a total count (e.g., **rRNA: 3**) without listing 16S vs 23S vs 5S in that summary.

To see which rRNAs were found: - Open the **.gff** file and search for `rRNA`, `16S`, `23S`, `5S` **OR** - Open the **.ffn** file and search for `ribosomal RNA`

5A. Paste the description line (header or attribute text) for the 16S rRNA feature:

gnl|Prokka|NCJPFIPA_56 barrnap:0.9 rRNA 51 1588 0 - . ID=NCJPFIPA_04563;Parent=NCJPFIPA_04563_gene;locus_tag=NCJPFIPA_04563;product=16S ribosomal

5B. Paste the description line (header or attribute text) for the 23S rRNA feature:

ID=NCJPFIPA_04556;Parent=NCJPFIPA_04556_gene;locus_tag=NCJPFIPA_04556;note=aligned only 53 percent of the 23S ribosomal RNA;product=23S ribosomal RNA (partial)

5C. Paste the description line (header or attribute text) for the 5S rRNA feature:

gnl|Prokka|NCJPFIPA_8 barrnap:0.9 rRNA 252 362 7.3e-11 + . ID=NCJPFIPA_02254;Parent=NCJPFIPA_02254_gene;locus_tag=NCJPFIPA_02254;product=5S ribosomal

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6) Hypothetical proteins — estimate and interpret

Hypothetical proteins typically appear as: - `product=hypothetical protein` in `.gff` - “hypothetical protein” in headers/annotations in `.faa`

6A. How did you estimate or count hypothetical proteins? (1–2 sentences)

Hit "command + F" on gff file and observed the count of the results. That count is my estimated number of hypothetical proteins.

6B. Estimated number of hypothetical proteins:

1098

6C. Estimated % hypothetical proteins (of CDS):

24.34%

6D. What does a high % hypothetical proteins mean (and NOT mean)? (2–3 sentences)

A high % hypothetical protein means that many of the predicted genes that encode proteins have an unknown function. However, this doesn't mean that the hypothetical proteins are fake or nonfunctional.

7) Extract 16S rRNA from the Prokka .ffn

Open the Prokka **.ffn** file. Search for a header/annotation indicating **16S ribosomal RNA**.

7A. Paste the FASTA header line for the 16S entry you used:

>NCJPFIPA_04563 16S ribosomal RNA

7B. Paste the full 16S rRNA sequence you used for BLAST (sequence only; you can keep line breaks):

TTGAAGAGTTTGATCATGGCTCAGATTGAACGCTGGCGGCAGGCCTAACACATGCAAGTC
GAACGGTAACGGGAAGCAGCTTGCTGCTTCGCTGACGAGTGGCGGACGGGTGAGTAATGT
CTGGGAAACTGCCTGATGGAGGGGGATAACTACTGGAAACGGTGGCTAATACCGCATAAC
GTCGCAAGACCAAAGAGGGGGACCTTCGGGCCTCTTGCCATCAGATGTGCCCAGATGGGA
TTAGCTTGTTGGTGAGGTAACGGCTCACCAAGGCGACGATCCCTAGCTGGTCTGAGAGGA
TGACCAGCCACACTGGAACTGAGACACGGTCCAGACTCCTACGGGAGGCAGCAGTGGGGA
ATATTGCACAATGGGCGCAAGCCTGATGCAGCCATGCCGCGTGTATGAAGAAGGCCTTCG
GGTTGTAAAGTACTTTCAGCGGGGAGGAAGGTGTTGTGGTTAATAACCGCAGCAATTGAC
GTTACCCGCAGAAGAAGCACCGGCTAACTCCGTGCCAGCAGCCGCGGTAATACGGAGGGT
GCAAGCGTTAATCGGAATTACTGGGCGTAAAGCGCACGCAGGCGGTCTGTCAAGTCGGAT

7C. In 1 sentence: why do we use 16S for a quick ID?

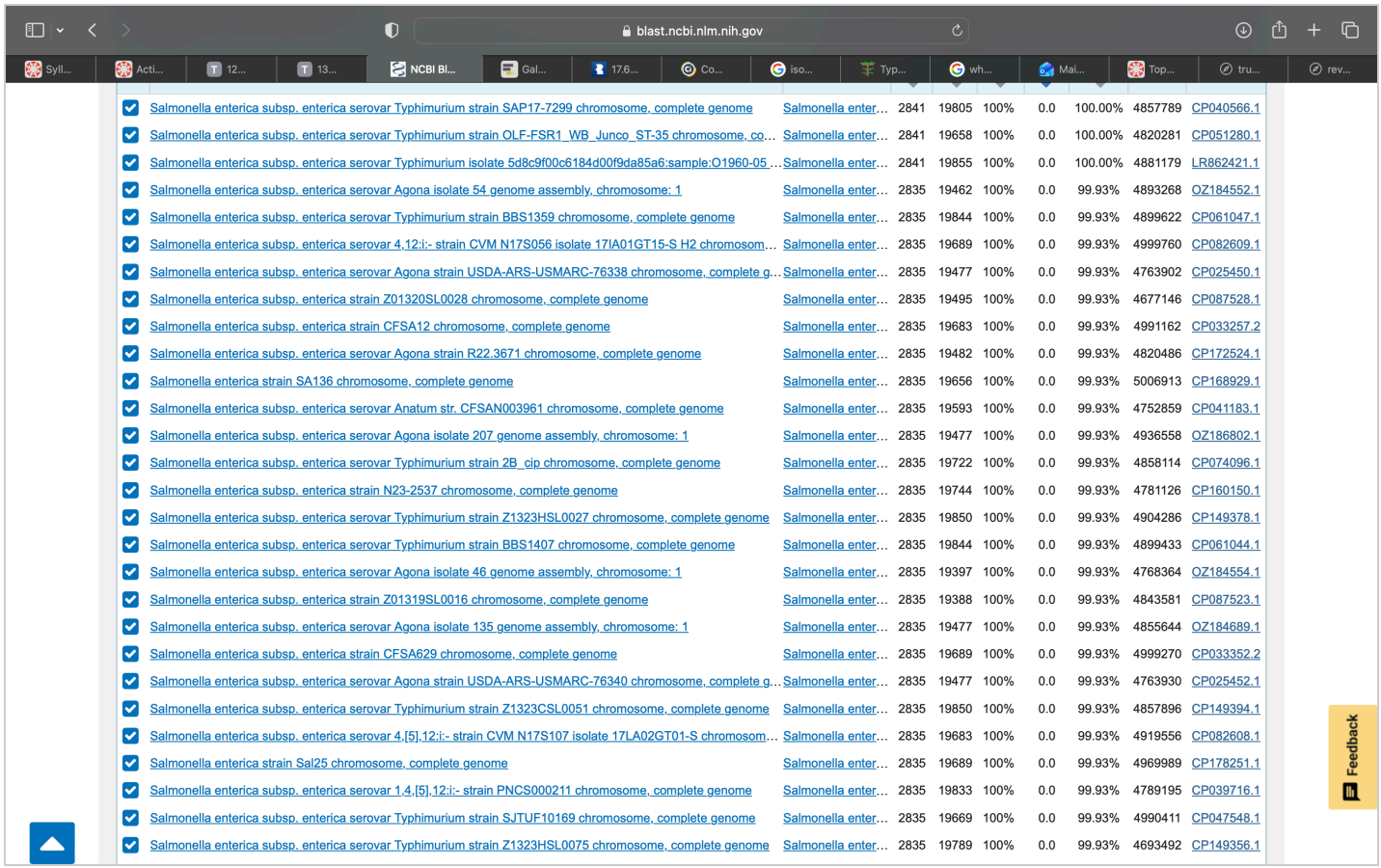
We use it because it is conserved in all bacteria and contains conserved regions that are distinguishable among different species.

8) 16S BLAST (species ID quick check)

BLAST your 16S sequence: 1. Go to NCBI BLAST 2. Choose **BLASTn** 3. Paste your 16S sequence 4. Run BLAST and choose a reasonable hit to interpret

Upload screenshot: your 16S BLAST results (Descriptions table or Alignment view)

Choose File Screensho….00.08 AM



Salmonella enterica subsp. enterica serovar Typhimurium strain SAP17-7299 chromosome, complete genome	Salmonella enter…	2841	19805	100%	0.0	100.00%	4857789	CP040566.1
Salmonella enterica subsp. enterica serovar Typhimurium strain OLF-FSR1_WB_Junco_ST-35 chromosome, co…	Salmonella enter…	2841	19658	100%	0.0	100.00%	4820281	CP051280.1
Salmonella enterica subsp. enterica serovar Typhimurium isolate 5d8c9f00c6184d00f9da85a6:sample:O1960-05 …	Salmonella enter…	2841	19855	100%	0.0	100.00%	4881179	LR862421.1
Salmonella enterica subsp. enterica serovar Agona isolate 54 genome assembly, chromosome: 1	Salmonella enter…	2835	19462	100%	0.0	99.93%	4893268	OZ184552.1
Salmonella enterica subsp. enterica serovar Typhimurium strain BBS1359 chromosome, complete genome	Salmonella enter…	2835	19844	100%	0.0	99.93%	4899622	CP061047.1
Salmonella enterica subsp. enterica serovar 4,12:i:- strain CVM N17S056 isolate 17IA01GT15-S H2 chromosom…	Salmonella enter…	2835	19689	100%	0.0	99.93%	4999760	CP082609.1
Salmonella enterica subsp. enterica serovar Agona strain USDA-ARS-USMARC-76338 chromosome, complete g…	Salmonella enter…	2835	19477	100%	0.0	99.93%	4763902	CP025450.1
Salmonella enterica subsp. enterica strain Z01320SL0028 chromosome, complete genome	Salmonella enter…	2835	19495	100%	0.0	99.93%	4677146	CP087528.1
Salmonella enterica subsp. enterica strain CFSA12 chromosome, complete genome	Salmonella enter…	2835	19683	100%	0.0	99.93%	4991162	CP033257.2
Salmonella enterica subsp. enterica serovar Agona strain R22.3671 chromosome, complete genome	Salmonella enter…	2835	19482	100%	0.0	99.93%	4820486	CP172524.1
Salmonella enterica strain SA136 chromosome, complete genome	Salmonella enter…	2835	19656	100%	0.0	99.93%	5006913	CP168929.1
Salmonella enterica subsp. enterica serovar Anatum str. CFSAN003961 chromosome, complete genome	Salmonella enter…	2835	19593	100%	0.0	99.93%	4752859	CP041183.1
Salmonella enterica subsp. enterica serovar Agona isolate 207 genome assembly, chromosome: 1	Salmonella enter…	2835	19477	100%	0.0	99.93%	4936558	OZ186802.1
Salmonella enterica subsp. enterica serovar Typhimurium strain 2B_cip chromosome, complete genome	Salmonella enter…	2835	19722	100%	0.0	99.93%	4858114	CP074096.1
Salmonella enterica subsp. enterica strain N23-2537 chromosome, complete genome	Salmonella enter…	2835	19744	100%	0.0	99.93%	4781126	CP160150.1
Salmonella enterica subsp. enterica serovar Typhimurium strain Z1323HSL0027 chromosome, complete genome	Salmonella enter…	2835	19850	100%	0.0	99.93%	4904286	CP149378.1
Salmonella enterica subsp. enterica serovar Typhimurium strain BBS1407 chromosome, complete genome	Salmonella enter…	2835	19844	100%	0.0	99.93%	4899433	CP061044.1
Salmonella enterica subsp. enterica serovar Agona isolate 46 genome assembly, chromosome: 1	Salmonella enter…	2835	19397	100%	0.0	99.93%	4768364	OZ184554.1
Salmonella enterica subsp. enterica strain Z01319SL0016 chromosome, complete genome	Salmonella enter…	2835	19388	100%	0.0	99.93%	4843581	CP087523.1
Salmonella enterica subsp. enterica serovar Agona isolate 135 genome assembly, chromosome: 1	Salmonella enter…	2835	19477	100%	0.0	99.93%	4855644	OZ184689.1
Salmonella enterica subsp. enterica strain CFSA629 chromosome, complete genome	Salmonella enter…	2835	19689	100%	0.0	99.93%	4999270	CP033352.2
Salmonella enterica subsp. enterica serovar Agona strain USDA-ARS-USMARC-76340 chromosome, complete g…	Salmonella enter…	2835	19477	100%	0.0	99.93%	4763930	CP025452.1
Salmonella enterica subsp. enterica serovar Typhimurium strain Z1323CSL0051 chromosome, complete genome	Salmonella enter…	2835	19850	100%	0.0	99.93%	4857896	CP149394.1
Salmonella enterica subsp. enterica serovar 4,[5],12:i:- strain CVM N17S107 isolate 17LA02GT01-S chromosom…	Salmonella enter…	2835	19683	100%	0.0	99.93%	4919556	CP082608.1
Salmonella enterica strain Sal25 chromosome, complete genome	Salmonella enter…	2835	19689	100%	0.0	99.93%	4969989	CP178251.1
Salmonella enterica subsp. enterica serovar 1,4,[5],12:i:- strain PNCS000211 chromosome, complete genome	Salmonella enter…	2835	19833	100%	0.0	99.93%	4789195	CP039716.1
Salmonella enterica subsp. enterica serovar Typhimurium strain SJTUF10169 chromosome, complete genome	Salmonella enter…	2835	19669	100%	0.0	99.93%	4990411	CP047548.1
Salmonella enterica subsp. enterica serovar Typhimurium strain Z1323HSL0075 chromosome, complete genome	Salmonella enter…	2835	19789	100%	0.0	99.93%	4693492	CP149356.1

8A. Top hit organism name (as shown):

Salmonella enterica Typhimurium strain

8B. % identity and query coverage for that hit:

100%

8C. Your best interpretation of species/genus (be cautious) (2–3 sentences):

The most likely species/genus is Salmonella. The top three, all at 100% identity and query coverage, suggest that it is likely Salmonella enterica.

8D. One reason 16S BLAST can be misleading or insufficient (1–2 sentences):

Many closely related bacterial species have very similar/identical 16S rRNA, which could be misleading for close matches.

9) ANI confirmation (EZBioCloud ANI)

After you have a likely species ID from **16S BLAST**, use ANI as a stronger genome-wide comparison.

You will use the **EZBioCloud ANI** tool here (open in a new tab): https://www.ezbiocloud.net/tools/ani (https://www.ezbiocloud.net/tools/ani)

What you will do 1. Download your **FILTERED contigs FASTA** (the same one you used for Prokka today). 2. Download **two reference genomes** (FASTA) from NCBI for *two candidate species* you want to compare against (often the top 16S BLAST hit and one close alternative). 3. Run ANI twice (your genome vs reference genome 1; your genome vs reference genome 2).

9A. Species/strain for comparison #1 (your chosen reference):

Salmonella typhimurium

9B. ANI for comparison #1 (%):

99.94%

9C. Species/strain for comparison #2 (your chosen reference):

Salmonella bongori

9D. ANI for comparison #2 (%):

89.82%

9E. Interpretation (2–3 sentences): Which comparison supports the strongest species-level match, and why?

Comparison

10) TYGS — what it provides

Submit your assembly to TYGS (Type Strain Genome Server). TYGS will: - Compare your genome to type strains - Provide **dDDH** (species boundary ~70%) - Produce a phylogenomic tree

10A. In your own words: what is TYGS doing that 16S BLAST is not? (2–4 sentences)

16s BLAST only looks for the rRNA which s shared among different species. TYGS mores directly compares the entire genome and finds the closest possible match.

10B. Record your closest type strain match (name) and dDDH value (if available today):

Salmonella typhimurium + 99.4%

10C. What does the 70% dDDH threshold mean? (1–2 sentences)

It is the threshold for determining confidence of a species match. If it is above the threshold it is usually the same species, but if it is less it usually means they are different species.

10D. If TYGS shows low branch support and offers a proteome-based tree, what does that imply? (2–3 sentences)

Low confidence in the matching of genomes, so they have to compare proteins which are more well-conserved. This implies that there may have been significant nucleotide changes, but not enough to change important protein function.

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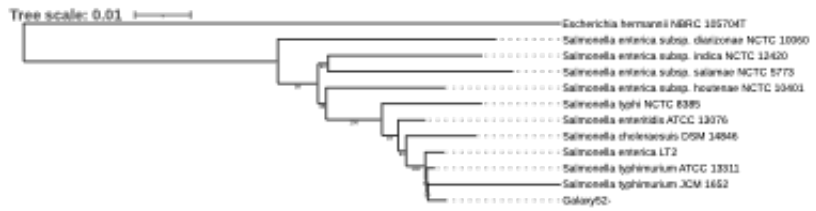
11) iTOL — visualize your TYGS tree

1. From TYGS, download the **tree file** (**.phy**)
2. Go to **iTOL** (Interactive Tree of Life)
3. Upload the tree
4. Adjust labels/formatting enough to be readable
5. Take a screenshot

Upload screenshot: iTOL tree visualization (from TYGS tree)

Choose File iTOL_salmo…atree.pdf



11A. What does the tree tell you that a single BLAST hit does not? (2–3 sentences)

It shows the relatedness on a "timeline-based" scale. BLAST just shows the relatedness based on the 16s rRNA whereas this puts that relatedness in the timeline.

11B. Identify one branch you trust and one you would be cautious about. Explain using support values if shown. (2–4 sentences)

I would trust the branch with the 100 bootstrap value just because of the amount it is rearranged and showed up in this format. I would be cautious about the one with 33 just because its lower, but that could just be because of recent mutations.

12) What to download and save for Git

Minimum recommended downloads for your local folder: - Prokka **.txt** summary - Prokka **.gff** - Prokka **.ffn** - Prokka **.faa** - CheckM2 results/report (table) - TYGS tree file (**.phy**) - Your PDF from this page (print-to-PDF)

12A. Write 4–6 lines of a "workflow note" you would put in `workflow_day13.md` (include tool versions when possible).

Parameters: Display bootstraps as text

Step 7.)
Average nucleotide identity
Tool: https://www.ezbiocloud.net/tools/ani
Version: Website

Export (PDF)

1. Upload the required screenshots above (BLAST + iTOL)
2. Click **Prepare submission preview**
3. Click **Download as PDF (Print)**

Submission preview

Setup
Assembly file: Sample1_spadescontigs.fasta
Local folder: Bioinformatics/biol4810/module02/annotation/

Prokka run
Version: Version 1.14.6+galaxy1
Settings: Force Genbank Compliance - YES

Outputs understanding
.gff vs .ffn vs .faa: .gff shows where genes and features are located on a genome by annotating where they are located (coordinates. .ffn contain the DNA sequences of genes. Finally, .faa shows the protein sequence that is predicted through the amino acid sequence.

Prokka summary (.txt)
Organism: Genus species strain
Contigs: 79 *Bases:* 4797060
CDS: 4507 *rRNA:* 3 *tRNA:* 62 *tmRNA:* 1
Sanity check: The CDS count falls within the range of a typical bacterial genome (1000-6000), and the genome size falls within the respective range for base pairs (1-6 million). With both of these taken into account, it appears that this looks like a typical bacterial genome.

rRNA evidence
16S description: gnl|Prokka|NCJPFIPA_56 barrnap:0.9 rRNA 51 1588 0 - . ID=NCJPFIPA_04563;Parent=NCJPFIPA_04563_gene;locus_tag=NCJPFIPA_04563;product=16S ribosomal RNA
23S description: ID=NCJPFIPA_04556;Parent=NCJPFIPA_04556_gene;locus_tag=NCJPFIPA_04556;note=aligned only 53 percent of the 23S ribosomal RNA;product=23S ribosomal RNA (partial)
5S description: gnl|Prokka|NCJPFIPA_8 barrnap:0.9 rRNA 252 362 7.3e-11 + . ID=NCJPFIPA_02254;Parent=NCJPFIPA_02254_gene;locus_tag=NCJPFIPA_02254;product=5S ribosomal RNA

CheckM2
CheckM2 version: (Galaxy Version 1.1.0+galaxy0)
Completeness: 98.96% *Contamination:* 0.1%
CheckM2 interpretation: I would trust this because of the high percentage of completeness and low percentage of contamination. Low contamination means that their sequence is mostly comprise from a single organism, and the high completeness indicates that the sequence contains most of the expected genes for the organism's genome.

Hypothetical proteins
How estimated: Hit "command + F" on gff file and observed the count of the results. That count is my estimated number of hypothetical proteins.
 # hypothetical: 1098 *% hypothetical:* 24.34%
Interpretation: A high % hypothetical protein means that many of the predicted genes that encode proteins have an unknown function. However, this doesn't mean that the hypothetical proteins are fake or nonfunctional.

16S extraction
16S header: >NCJPFIPA_04563 16S ribosomal RNA
16S sequence:

```
TTGAAGAGTTTGATCATGGCTCAGATTGAACGCTGGCGGCAGGCCTAACACATGCAAGTC
GAACGGTAACGGGAAGCAGCTTGCTGCTTCGCTGACGAGTGGCGGACGGGTGAGTAATGT
CTGGGAAACTGCCTGATGGAGGGGGATAACTACTGGAAACGGTGGCTAATACCGCATAAC
GTCGCAAGACCAAAGAGGGGGACCTTCGGGCCTCTTGCCATCAGATGTGCCCAGATGGGA
TTAGCTTGTTGGTGAGGTAACGGCTCACCAAGGCGACGATCCCTAGCTGGTCTGAGAGGA
TGACCAGCCACACTGGAACTGAGACACGGTCCAGACTCCTACGGGAGGCAGCAGTGGGGA
ATATTGCACAATGGGCGCAAGCCTGATGCAGCCATGCCGCGTGTATGAAGAAGGCCTTCG
GGTTGTAAAGTACTTTCAGCGGGGAGGAAGGTGTTGTGGTTAATAACCGCAGCAATTGAC
GTTACCCGCAGAAGAAGCACCGGCTAACTCCGTGCCAGCAGCCGCGGTAATACGGAGGGT
GCAAGCGTTAATCGGAATTACTGGGCGTAAAGCGCACGCAGGCGGTCTGTCAAGTCGGAT
GTGAAATCCCCGGGCTCAACCTGGGAACTGCATTCGAAACTGGCAGGCTTGAGTCTTGTA
GAGGGGGGTAGAATTCCAGGTGTAGCGGTGAAATGCGTAGAGATCTGGAGGAATACCGGT
GGCGAAGGCGGCCCCCTGGACAAAGACTGACGCTCAGGTGCGAAAGCGTGGGGAGCAAAC
AGGATTAGATACCCTGGTAGTCCACGCCGTAAACGATGTCTACTTGGAGGTTGTGCCCTT
GAGGCGTGGCTTCCGGAGCTAACGCGTTAAGTAGACCGCCTGGGGAGTACGGCCGCAAGG
TTAAAACTCAAATGAATTGACGGGGGCCCGCACAAGCGGTGGAGCATGTGGTTTAATTCG
ATGCAACGCGAAGAACCTTACCTGGTCTTGACATCCACAGAAGAATCCAGAGATGGATTT
GTGCCTTCGGGAACTGTGAGACAGGTGCTGCATGGCTGTCGTCAGCTCGTGTTGTGAAAT
GTTGGGTTAAGTCCCGCAACGAGCGCAACCCTTATCCTTTGTTGCCAGCGATTAGGTCGG
GAACTCAAAGGAGACTGCCAGTGATAAACTGGAGGAAGGTGGGGATGACGTCAAGTCATC
ATGGCCCTTACGACCAGGGCTACACACGTGCTACAATGGCGCATACAAAGAGAAGCGACC
TCGCGAGAGCAAGCGGACCTCATAAAGTGCGTCGTAGTCCGGATTGGAGTCTGCAACTCG
ACTCCATGAAGTCGGAATCGCTAGTAATCGTGGATCAGAATGCCACGGTGAATACGTTCC
CGGGCCTTGTACACACCGCCCGTCACACCATGGGAGTGGGTTGCAAAAGAAGTAGGTAGC
TTAACCTTCGGGAGGGCGCTTACCACTTTGTGATTCATGACTGGGGTGAAGTCGTAACAA
GGTAACCGTAGGGGAACCTGCGGTTGGATCACCTCCTT
```

Why 16S: We use it because it is conserved in all bacteria and contains conserved regions that are distinguishable among different species.

16S BLAST

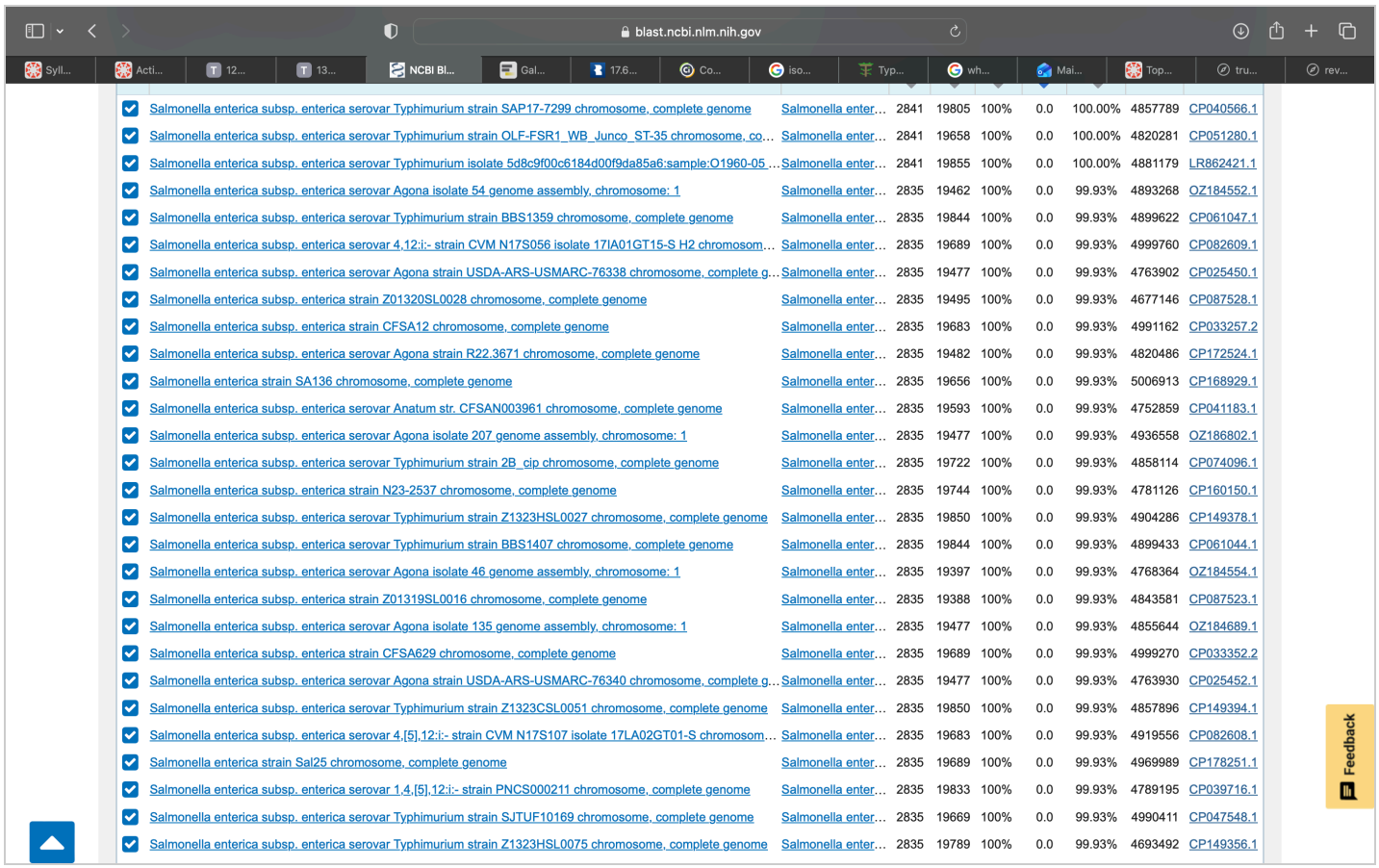
Top hit: Salmonella enterica Typhimurium strain

Metrics: 100%

Interpretation: The most likely species/genus is Salmonella. The top three, all at 100% identity and query coverage, suggest that it is likely Salmonella enterica.

Limitation: Many closely related bacterial species have very similar/identical 16S rRNA, which could be misleading for close matches.

BLAST screenshot



Salmonella enterica subsp. enterica serovar Typhimurium strain SAP17-7299 chromosome, complete genome	Salmonella enter...	2841	19805	100%	0.0	100.00%	4857789	CP040566.1
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Salmonella enterica subsp. enterica serovar Typhimurium isolate 5d8c9f00c6184d00f9da85a6:sample:O1960-05 ...	Salmonella enter...	2841	19855	100%	0.0	100.00%	4881179	LR862421.1
Salmonella enterica subsp. enterica serovar Agona isolate 54 genome assembly, chromosome: 1	Salmonella enter...	2835	19462	100%	0.0	99.93%	4893268	OZ184552.1
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Salmonella enterica subsp. enterica serovar 4,12:i:- strain CVM N17S056 isolate 17IA01GT15-S H2 chromosom...	Salmonella enter...	2835	19689	100%	0.0	99.93%	4999760	CP082609.1
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Salmonella enterica subsp. enterica serovar Typhimurium strain 2B_cip chromosome, complete genome	Salmonella enter...	2835	19722	100%	0.0	99.93%	4858114	CP074096.1
Salmonella enterica subsp. enterica strain N23-2537 chromosome, complete genome	Salmonella enter...	2835	19744	100%	0.0	99.93%	4781126	CP160150.1
Salmonella enterica subsp. enterica serovar Typhimurium strain Z1323HSL0027 chromosome, complete genome	Salmonella enter...	2835	19850	100%	0.0	99.93%	4904286	CP149378.1
Salmonella enterica subsp. enterica serovar Typhimurium strain BBS1407 chromosome, complete genome	Salmonella enter...	2835	19844	100%	0.0	99.93%	4899433	CP061044.1
Salmonella enterica subsp. enterica serovar Agona isolate 46 genome assembly, chromosome: 1	Salmonella enter...	2835	19397	100%	0.0	99.93%	4768364	OZ184554.1
Salmonella enterica subsp. enterica strain Z01319SL0016 chromosome, complete genome	Salmonella enter...	2835	19388	100%	0.0	99.93%	4843581	CP087523.1
Salmonella enterica subsp. enterica serovar Agona isolate 135 genome assembly, chromosome: 1	Salmonella enter...	2835	19477	100%	0.0	99.93%	4855644	OZ184689.1
Salmonella enterica subsp. enterica strain CFSA629 chromosome, complete genome	Salmonella enter...	2835	19689	100%	0.0	99.93%	4999270	CP033352.2
Salmonella enterica subsp. enterica serovar Agona strain USDA-ARS-USMARC-76340 chromosome, complete g...	Salmonella enter...	2835	19477	100%	0.0	99.93%	4763930	CP025452.1
Salmonella enterica subsp. enterica serovar Typhimurium strain Z1323CSL0051 chromosome, complete genome	Salmonella enter...	2835	19850	100%	0.0	99.93%	4857896	CP149394.1
Salmonella enterica subsp. enterica serovar 4,[5],12:i:- strain CVM N17S107 isolate 17LA02GT01-S chromosom...	Salmonella enter...	2835	19683	100%	0.0	99.93%	4919556	CP082608.1
Salmonella enterica strain Sal25 chromosome, complete genome	Salmonella enter...	2835	19689	100%	0.0	99.93%	4969989	CP178251.1
Salmonella enterica subsp. enterica serovar 1,4,[5],12:i:- strain PNCS000211 chromosome, complete genome	Salmonella enter...	2835	19833	100%	0.0	99.93%	4789195	CP039716.1
Salmonella enterica subsp. enterica serovar Typhimurium strain SJTUF10169 chromosome, complete genome	Salmonella enter...	2835	19669	100%	0.0	99.93%	4990411	CP047548.1
Salmonella enterica subsp. enterica serovar Typhimurium strain Z1323HSL0075 chromosome, complete genome	Salmonella enter...	2835	19789	100%	0.0	99.93%	4693492	CP149356.1

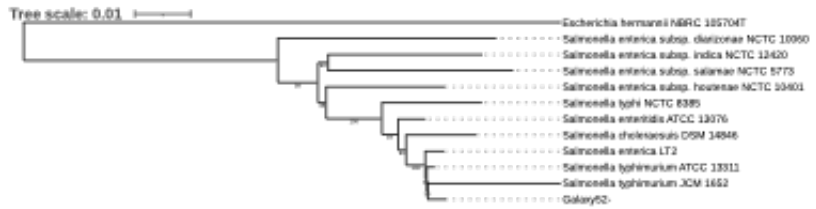
TYGS

TYGS vs 16S: 16s BLAST only looks for the rRNA which s shared among different species. TYGS mores directly compares the entire genome and finds the closest possible match.
Closest match + dDDH: Salmonella typhimurium + 99.4%
70% threshold: It is the threshold for determining confidence of a species match. If it is above the threshold it is usually the same species, but if it is less it usually means they are different species.
Proteome tree option (low support): Low confidence in the matching of genomes, so they have to compare proteins which are more well-conserved. This implies that there may have been significant nucleotide changes, but not enough to change important protein function.

iTOL tree

Tree adds value because: It shows the relatedness on a "timeline-based" scale. BLAST just shows the relatedness based on the 16s rRNA whereas this puts that relatedness in the timeline.
Support/trust: I would trust the branch with the 100 bootstrap value just because of the amount it is rearranged and showed up in this format. I would be cautious about the one with 33 just because its lower, but that could just be because of recent mutations.

iTOL tree screenshot



Workflow note (for Git)

```
Annotation pipeline 3.10.26

Step1.)
Filtering contains shorter than 500 BPs by using Filter FASTA (Galaxy Version 2.3)
Parameters were on sequence length minimum length BP

Step2.)
Quality assessment using CheckM2
Tool: CheckM2
Version: Galaxy Version 1.10+galaxy0
Parameters: Model Options - Output Quality prediction for both models for each
genome

Step 3.)
Genome annotation
Tools: Prokka
Version: Galaxy Version 1.14.6+galaxy1
Parameters: Force Genbank Compliance - YES

Step 4.)
Species ID
Tool: BLAST
Version: BLASTn (web)
```

```
Parameter: rRNA/ITS database

Step 5.)
Build phylogenetic tree
Tool: Type Strain Genome Server
Version: https://tygs.dsmz.de/ (web)
Parameters: whole Proteome Tree

Step 6.)
Visualize the phylogenetic tree
Tool: Interactive Tree of Life (iTOL)
Version: v7 (web)
Parameters: Display bootstraps as text

Step 7.)
Average nucleotide identity
Tool: https://www.ezbiocloud.net/tools/ani
Version: Website
Parameters: From NCBI genome database
(https://www.ncbi.nlm.nih.gov/datasets/genome/), downloaded the FASTA file for
Salmonella typhimurium (closest hit on 16S BLAST) and Salmonella bongori (more
distant relative). Compared filteredcontigs.fasta to the FASTA files from
typhimurium and S. bongori

Step 8.)
Upload on GitHub steps on command line:
a.) cd into directory (can drag and drop) (module02 in this case)
b.) git status to confirm GitHub directory
c.) git add module02/annotation/
d.) git status to confirm in git repository
e.) git commit -m “message”
f.) git push
```